

Highlights

- PRR frames prediction as candidate generation rather than reserved population insertion.
- Predicted, memory, transfer, elite, and diversity candidates compete under current-state selection before admission.
- The method is positioned as a conservative and interpretable evolutionary soft-computing mechanism for rapid recovery under variable prediction reliability.
- Ablation reports no-memory, no-elite, no-prediction, no-diversity, and direct-insertion boundary cases.
- Scenario-level ranks, evaluation-count audit, direct recovery diagnostics, component-risk ablation, quota sensitivity, and prediction-noise stress tests are reported.